GATE
CS & DA

Artificial Intelligence

Uninformed search

Discussion Notes

DPP- 01

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[MCQ]

#Q. The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below ?

A. MNOPQR

B. NQMPOR

C. QMNROP

D. POQNMR

MNR

NMQO

POQNMR

[MCQ]

#Q. The time and space complexity of BFS is (For time and space complexity problems consider b as branching factor and d as depth of the search tree.)

(b^d)

A. O(b^d+1) and O(b^d+1)

B. O(b2) and O(d2)

C. O(d2) and O(b2)

D. O(d2) and O(d2)

[MCQ]

#Q. Consider the following graph:

Among the following sequences:

I → abeghf

II → abfehg

III → abfhge

IV → afghbe

Which are the depth-first traversals of the above graph ?

abe
abf
abf

A. I, II and IV only

B. I and IV only

C. II, III and IV only

D. I, III and IV only

1) Suppose we do Iterative Deepening Search (IDS) but instead of increasing depths as 1,2,3,4… we increase it as 1,2,4,8,16… i.e. double the depth instead of increasing it by

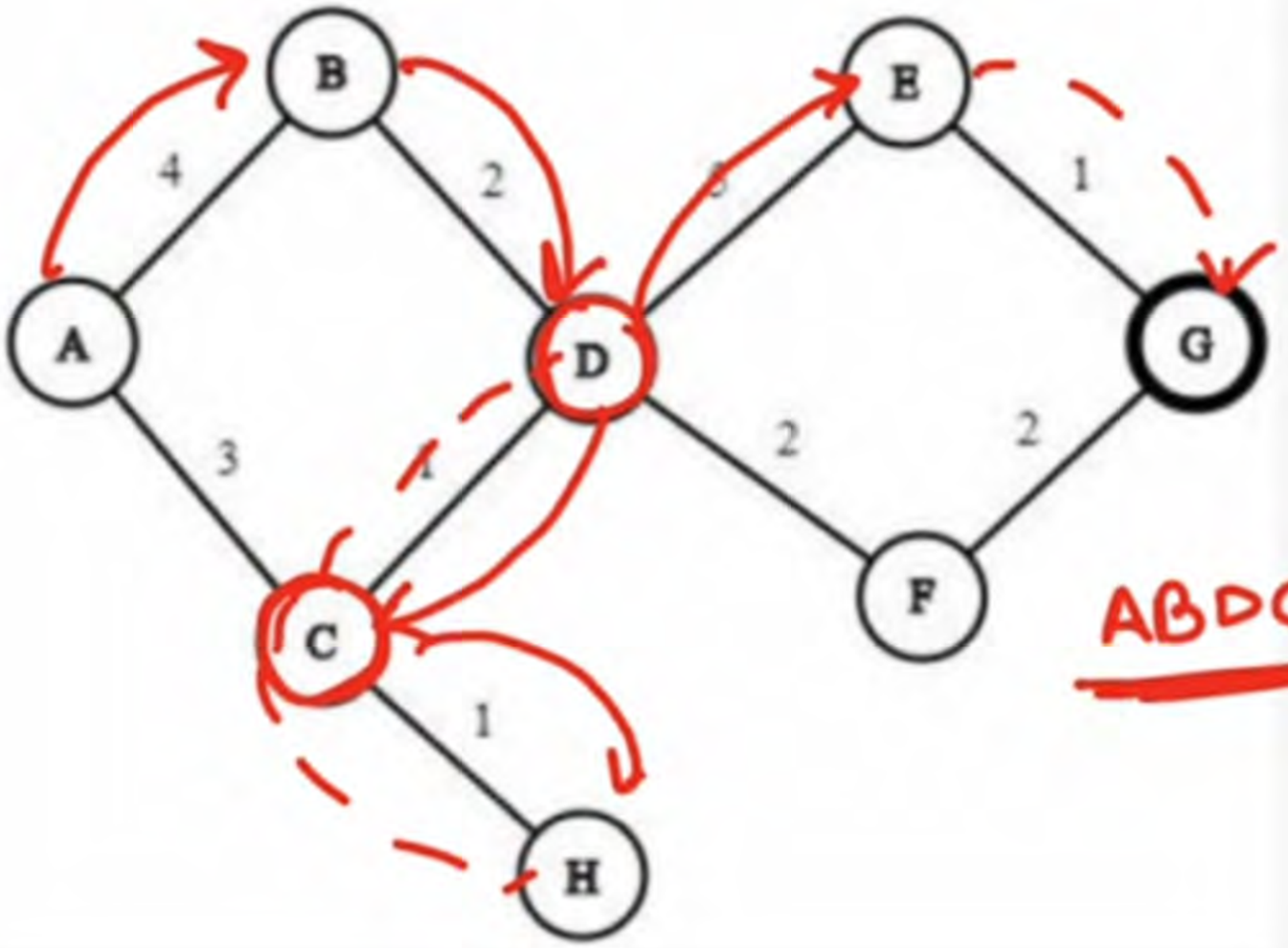
1. This modified version of IDS is

- [ ] Optimal but not complete
- [x] Complete but not optimal
- [ ] More efficient than standard IDS in the worst case
- [ ] Less efficient than standard IDS in the worst case

2) Suppose there is only one goal state and each step cost is k (k>0). Which of the following search algorithm(s) will return the optimal path?

- [x] Breadth-First Search
- [ ] Depth First Search
- [x] Uniform Cost Search
- [x] Iterative Deepening Search

3) Consider the following graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Find the path to the goal found by Depth First Search with full duplicate detection, which explores children in lexicographical order. (Write answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD)



ABDCHEGI

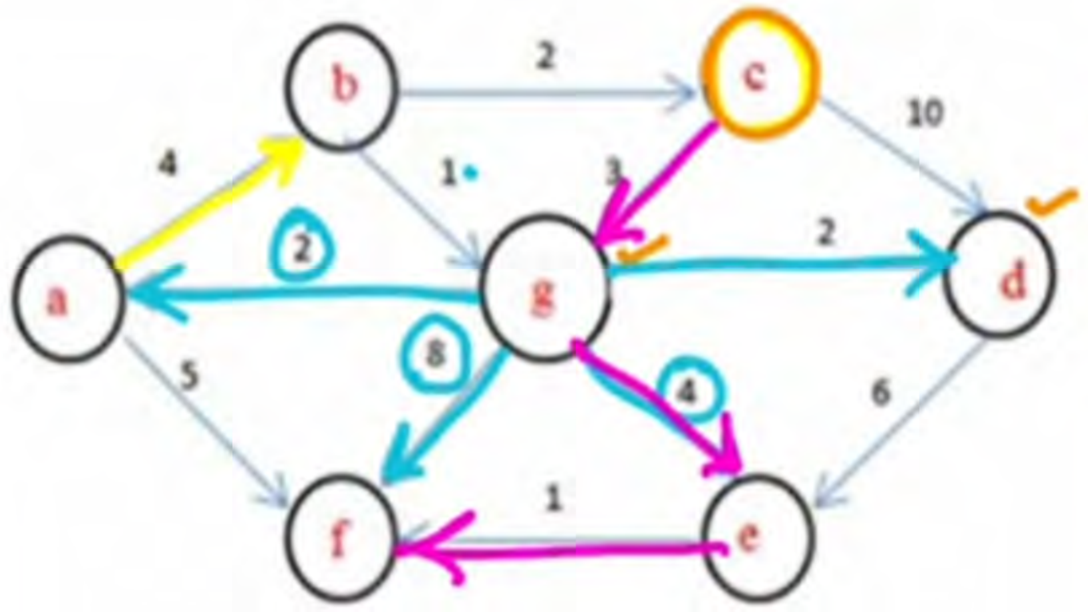
A C F G

10) Consider the depth-limited search algorithm with depth limit f. If the only goal state exists at depth d then which of the following statement(s) is/are correct?

if f > d

- If f > d then it is complete but not optimal.
- If f > d then it is not complete and not optimal.
- If f < d then it is complete but not optimal.
- If f < d then it is not complete and not optimal.

10) Consider the following graph:



What is the minimum cost to reach vertex f starting from vertex c?

	c	g	a	d	b	e
a	∞	5	5	5	5	5
b	∞	∞	9	9	9	9
c	0	0	0	0	0	0
d	10	5	5	5	5	5
e	∞	7	7	7	7	7
f	∞	11	11	11	11	8
g	3	3	3	3	3	

8

5) Consider a 2-D grid where each coordinate represents a location. A robot is standing at (0, 0) and the goal exists at (2, 2). In each move, the robot can either move up, down, left, or right with equal probability. What is the probability that the robot will reach the goal in 4 moves?

Give the answer rounded to three decimal places.

abefc
adgebcf
adbc

#Q. Consider the following sequence of nodes for the undirected graph given below:

1. abefdgc
2. abefcgd
3. adgebcf
4. adbcgef

A Depth First Search (DFS) is started at node 'a'. The nodes are listed in the order they are first visited. Which of the above is/are possible output(s)?

A. 1 and 3 only

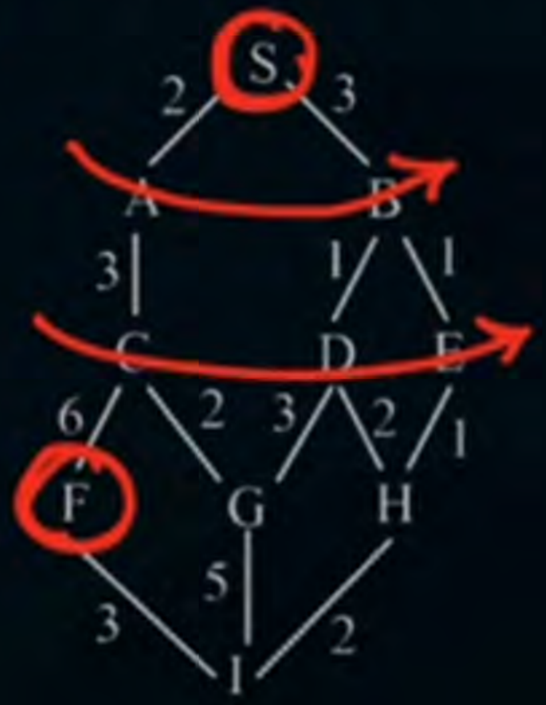
B. 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2 and 3 only

[MCQ]

#Q. Consider the graph below with F,G,H as the goal nodesBy applying BFS, the algorithm will end at which Goal Node alphabetical



A F

B G

C H

D I

Activate W
Go to Settings

[MCQ]

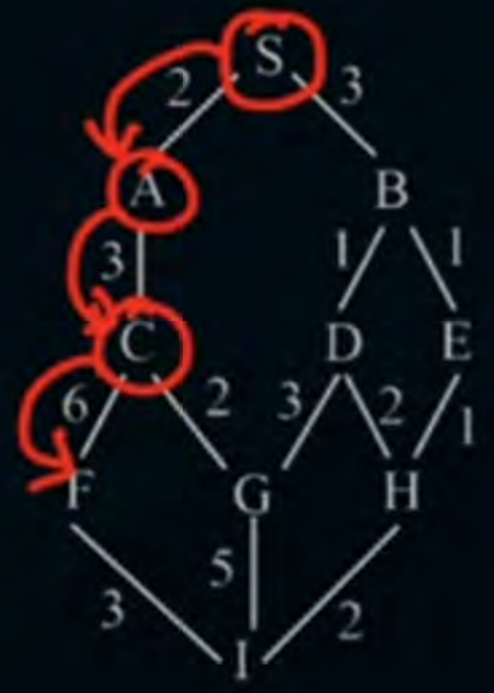
#Q. Consider the graph below with F,G,H by applying DFS, the algorithm will end at which Goal Node

A. F

B. G

C. H

D. I



[MCQ]

#Q. Consider the graph below with F,G,H by applying UCS, the algorithm will end at which Goal Node

A. F

B. G

C. H

D. I

S	0	0	0	0	0
A	2	2	2	2	2
B	3	3	3	3	3
C	∞	5	5	5	5 ←
D	∞	∞	4	4	4
E	∞	∞	4	4	4
F	∞	∞	∞	∞	∞
G	∞	∞	∞	7	7
H	∞	∞	∞	6	5 ←
I	∞	∞	∞	∞	∞

[MCQ]

#Q. Consider the graph below with goal nodes R and T ,by applying BFS, the algorithm will end at which Goal Node

A. P

B. Q

C. R ✓

D. T

Activate
Go to Setti

[MCQ]

#Q. Consider the graph below with goal nodes R and T by applying DFS, the algorithm will end at which Goal Node

A. P

B. Q

C. R

D. T

Activate W

[MCQ]

#Q. Consider the graph below with goal nodes R and T by applying UCS, the algorithm will end at which Goal Node

S	0	0	0
P	1	1	1
Q	∞	3	3
R	∞	5	5 ✓
T	∞	∞	4 ✓
U	∞	∞	∞
V	∞	∞	∞
W	∞	∞	∞

A P

B Q

C R

D T

#Q. In a graph where the goal is to reach from node S to node G, the following edges and costs are given:

- S to A. cost 1
- S to B: cost 2
- a to C: cost 3
- B to C: cost 1
- C to G. cost 2

S	0	0	0	0
A	1	1	1	1
B	2	2	2	2
C	∞	4	3	3
G	∞	∞	∞	5

Using Uniform Cost Search, what is the total cost of the path found by UCS if it finds the shortest path to G?

A 4

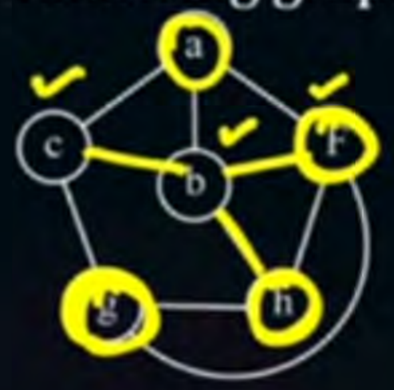
B 5

C 6

D 7

abfhgc
abcghf
afghbc

#Q. Consider the following graph:



For the graph; the following sequences of depth first search (DFS) are given

(A) abcghf
(B) abfchg
(C) abfhgc
(D) afghbc

Which of the following is correct?

A (A), (B) and (D) only

B (A), (B), (C) and (D)

C (B), (C) and (D) only

D (A), (C) and (D) only

#Q. Consider the following graph.

abfhge

abeghf

afghbe

Among the following sequences

I. a b e g h f

II. a b f e h g

III. a b f h g e

IV. a f g h b e

Which are depth first traversals of the above graph?

A I, II, and IV only

B I and IV only

C II, III, and IV only

D I, III, and IV only

Activate W
Go to Setting

THANK - YOU